

Reproduction of "Good, Bad, and Ugly Colonial Activities: Do They Matter for Economic Development?" by Miriam Bruhn and Francisco A. Gallego (2012)

Abstract

Bruhn and Gallego (2012) examined the long-term effects of colonial economic activities on modern regional development in Latin America. They classified historical activities into "good," "bad," "ugly," or "none," and used variation across regions within countries to estimate their relationship with present-day GDP per capita and poverty rates. Controlling for pre colonial population density, geography, and country fixed effects, they found that regions with bad colonial activities exhibit significantly lower GDP and higher poverty present-day. We have conducted a reproduction of the original analysis using the same dataset and statistical procedures. While omitting the "ugly" category due to missing data, our results confirm the sign, magnitude, and statistical significance of the original estimates regarding the negative effects of bad colonial legacies on modern development outcomes.

1 Introduction

The paper written by Miriam Bruhn and Francisco A. Gallego (2012) examines the historical roots of regional inequality in economic development across Latin America. In particular, the study examines whether different types of colonial economic activities (specifically those that relied on economies of scale and labor coercion) have long-term effects on present-day outcomes, such as income and poverty. Using data at the subnational level, the authors classify regions into four categories based on their dominant colonial activities: “good,” “bad,” “ugly,” or “none.” These classifications reflect differing institutional legacies, with “bad” and “ugly” activities associated with high social inequality and exploitation, and “good” associated with more inclusive, small-scale economic structures. The conclusions of the study suggest that regions historically engaged in bad colonial activities experience significantly lower GDP per capita and higher poverty rates present-day compared to regions with good or no colonial activities. Although evidence on “Ugly” activities is less consistent, the study provides strong empirical support for the idea that colonial institutions play a strong role in shaping modern development outcomes. We conducted a reproduction of this study using the available dataset and the original statistical procedures. Our replication successfully reproduces the findings regarding the long-run effects of bad colonial activities on both income and poverty. However, the available dataset does not include observations for regions classified as having “Ugly” colonial activities, resulting in estimates that differ slightly from those presented in the original paper. Nevertheless, the sign, magnitude, and statistical significance of the main coefficients are consistent, supporting the authors’ central claim that colonial economic structures continue to shape development today.

2 Reproducibility

There weren’t any coding errors found from the original STATA code by the authors. However, when trying to reproduce the results, we found that data corresponding

to “Ugly” colonial activities was not included in the public datasets. As a result, our reproduced estimates differ from the original paper, particularly in tables where “Ugly” activities were included as a separate category. The direction, magnitude, and statistical significance of our results remain broadly consistent with the original findings.

2.1 Challenges

2.1.1 Table 2: We were able to reproduce it almost identically, with the exception of the Ugly activities outcome variable (eugly), as this data is not included.

2.1.2 Figure 2: We were able to reproduce the choropleth map for the US almost identically, with the exception of the states of California, Massachusetts, and Rhode Island being classified in the shade for “Bad” rather than “Ugly,” as this data is not included.

2.1.3 Table 5, 6, 7, 8: The regressions match those in the original paper, with some minor differences in the coefficients, which will be discussed in the next section. These regressions omit the Ugly activities variable (eugly), as this data is not included.

3 Replication

This section presents our replication of Bruhn and Gallego (2012), focusing on reproducing descriptive and regression-based results from Tables 1, 2, 5, 6, 7, 8, and Figure 2. We describe the analytical approach, coding decisions, and any differences due to data availability.

3.1 Descriptive Tables

3.1.1 Table 1: Table 1 reports the regional PPP GDP per capita in the sub-national regions of each country. The key takeaway is that there is a substantial amount of inequality in income levels not just across countries but within them.

This descriptive table shows that regional disparities are large enough to justify subnational regression analysis. Our successful replication confirms that regional inequality is significant and comparable to cross-country inequality. We grouped the dataset by country, calculated summary statistics, and computed the ratio of maximum to minimum GDP per capita per country to reflect income dispersion. Rounding was applied to match formatting from the original paper.

3.1.2 Table 2: Table 2 provides descriptive statistics for the key variables throughout the paper, including modern development indicators (log GDP per capita, poverty rate, health index) as well as historical variables (colonial activity dummies, pre colonial population density). The means, standard deviations, and ranges we computed match the original table, with the exception of the statistics for the missing “ugly” data. Accurately replicating Table 2 is essential for interpreting the regression results later in the paper in understanding variation in data and distributions.

3.2 Visualizations

3.2.1 Figure 2: Figure 2 presents choropleth maps of countries in the Americas, shading each region according to the dominant type of colonial activity (Good, Bad, Ugly, or None). This spatial variation shows that colonial strategies varied geographically, and can help explain why some regions are wealthier or poorer today. The visual provides the foundation for subsequent regression analysis. We successfully reproduced the choropleth map for the US almost identically to the paper. However, the states California, Massachusetts, and Rhode Island were shaded as having “Bad” colonial activities rather than “Ugly,” due to the absence of “Ugly” classification data. This resulted in slightly different color shading compared to the original paper, as the visualization cannot exactly match the original without the full data. While the general spatial patterns and purpose of the map visualization remain intact, the exclusion of “Ugly” activities may not fully capture the diversity of colonial institutions and could bias interpretations toward Bad/Good distinc-

tion only; however, the figure still effectively conveys the idea that colonial legacies are unevenly distributed within countries. We used the colonial activity dummy variables available in the dataset to generate a regional classification.

3.3 Regressions

3.3.1 Table 5: Below is the regressions specification for equation (1) in Table 5:

$$\text{lyppp}_i = \beta_0 + \beta_1 \text{egood}_i + \beta_2 \text{ebad}_i + \varepsilon_i \quad (1)$$

This includes the egood and ebad dummy variables to test whether these colonial activity types are associated with modern income levels. Note that this omits eugly, since it is missing from the data. In order to properly reproduce the result, we added country fixed effects.

$$\text{lyppp}_i = \beta_0 + \beta_1 \text{egood}_i + \beta_2 \text{ebad}_i + \gamma_i + \varepsilon_i \quad (2)$$

In this specification, γ_i represents the country fixed effects. We found a large, negative, and statistically significant coefficient on ebad, consistent with the original table. This suggests that regions with bad colonial activities have substantially lower income levels today. The “ugly” category was not included due to missing data; however, statistically significant results matched those of the original article, with results being in the same direction with similar values for R^2 . Although the size of the coefficients differed, this is likely due to omitted variable bias from the missing data.

We replicated this regression to include the necessary variables for each of the 6 regressions included in Table 5.

3.3.2 Table 6: Below is the regressions specification for equation (1) in Table 6:

$$\text{lpoverity}_i = \beta_0 + \beta_1 \text{egood}_i + \beta_2 \text{ebad}_i + \varepsilon_i \quad (3)$$

This is the first specification from Table 6, where the dependent variable is the log of the poverty rate, lpoverty_i . It includes dummy variables for good and bad colonial activities and omits "ugly" due to missing data. In order to properly reproduce the result, we added country fixed effects.

$$\text{lpoverty}_i = \beta_0 + \beta_1 \text{egood}_i + \beta_2 \text{ebad}_i + \gamma_i + \varepsilon_i \quad (4)$$

In this specification, country fixed effects γ_i are added to account for national-level factors. Statistically significant results matched those of the original article, with results being in the same direction with similar values for R^2 . Although the size of the coefficients differed, this is likely due to omitted variable bias from the missing data. The coefficient on ebad was positive and statistically significant in both specifications, indicating that regions with bad colonial activities tend to have higher poverty rates today.

We replicated this regression to include the necessary variables for each of the 5 regressions included in Table 6.

3.3.3 Table 7: Baseline:

$$\begin{aligned} \text{Development}_{it} = & \beta_0 + \beta_1 \text{Good}_i + \beta_2 (\text{Good}_i \times \text{Post}_t) + \beta_3 \text{Bad}_i + \beta_4 (\text{Bad}_i \times \text{Post}_t) \\ & + \beta_5 \log(\text{PopDensity}_i) + \beta_6 (\log(\text{PopDensity}_i) \times \text{Post}_t) + \varepsilon_{it} \end{aligned} \quad (5)$$

This is the basic DiD setup for our regression. It includes variables that represent economic differences when comparing the precolonial and postcolonial periods, with interaction terms for colonial activity types (Good x Post, Bad x Post) and for precolonial population density. This tests whether development trajectories diverged after colonization based on type of colonial activity. Once again, 'eugly' is omitted from our regression due to not having that data.

With uninteracted post, and country FE, climate and geography controls:

$$\begin{aligned} \text{Development}_{it} = & \beta_0 + \beta_1 \text{Good}_i + \beta_2 (\text{Good}_i \times \text{Post}_t) + \beta_3 \text{Bad}_i + \beta_4 (\text{Bad}_i \times \text{Post}_t) \\ & + \beta_5 \log(\text{PopDensity}_i) + \beta_6 (\log(\text{PopDensity}_i) \times \text{Post}_t) + \beta_7 \text{Post}_t \\ & + \gamma_i + \mathbf{X}'_i \delta + \theta_t + \varepsilon_{it} \end{aligned} \quad (6)$$

In this specification, we control for a selection of significant variables to improve the model. By adding country-fixed effects γ_i along with time effect θ_t and climate and geography controls $\mathbf{X}'_i \delta$, we isolate the good vs bad activities during pre vs postcolonial periods. Results from our table matched the direction of all those in the original article, with the coefficient values differing due to the omission of 'eugly' in the dataset given.

3.3.4 Table 8: There were only two specifications for this table – the point was to analyze the correlation between the log of precolonial population density and the log of PPP GDP per capita for countries with (egood or ebad) colonial activity compared to those without (enone). In both cases, we accounted for country fixed effects (γ_i) as well as controlling for climate and geography ($\mathbf{X}'_i \delta$).

$$\text{lyppp}_y = \beta_0 + \beta_1 \log(\text{PopDensity}_y) + \gamma_i + \mathbf{X}'_i \delta + \varepsilon_i \quad (7)$$

The first specification is the case where we filtered for rows with colonial activity, so where there were positive values for 'egood' or 'ebad'. We were able to replicate the paper's negative coefficient, the same outcome of the p-test (fail to reject null hypothesis), and a comparable R^2 value.

$$\text{lyppp}_n = \beta_0 + \beta_1 \log(\text{PopDensity}_n) + \gamma_i + \mathbf{X}'_i \delta + \varepsilon_i \quad (8)$$

The second specification is the case where we filtered for rows with no colonial activity, so where 'enone' had a positive value. The regression was performed in the same way with just separate data. Our coefficient for this data was greater than

what the paper calculated, with a significantly higher p-value as well, even though the R^2 value is around the same. According to our results, if we use a 0.05 p-value cutoff, we reject the null but the paper would fail to reject which means they find that there should be some relationship between the log of PPP GDP per capita and log of precolonial population density.

4 Conclusion

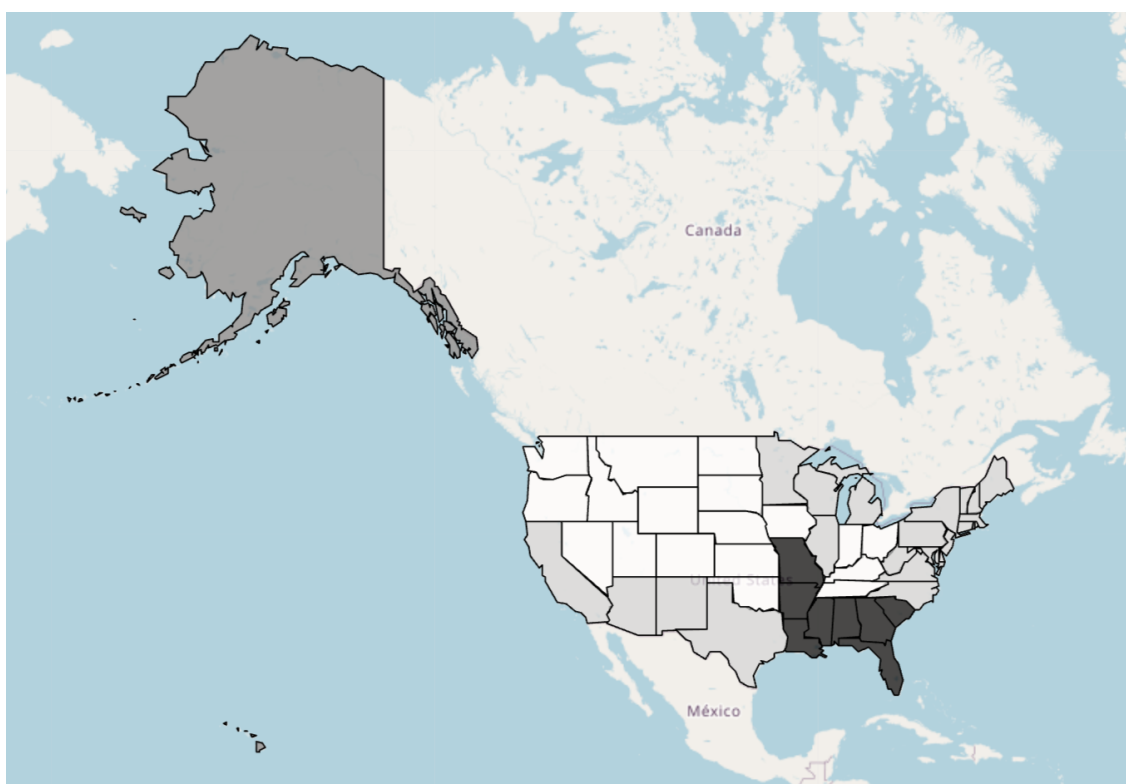
Overall, the reproduction was successful but posed challenges primarily due to limitations in the available dataset. In particular, the absence of regional data for “Ugly” colonial activities meant that we were unable to fully replicate figures and regressions that involved all four colonial categories. As a result, our analysis relied only on comparisons between “Good,” “Bad,” and “None” for activities. This did not prevent us from reproducing the paper’s central findings, but did limit our ability to evaluate the full spectrum of colonial legacies outlined by the authors. Despite this limitation, we were able to replicate several of the key tables and regression models using Python packages such as statsmodels, which allowed us to estimate OLS regressions with country fixed effects. The evidence we reproduced closely supports the original paper’s core claims that regions with bad colonial activities exhibit significantly lower levels of economic development today, while regions with good or no colonial activities tend to fare better. We also confirmed a negative association between pre colonial population density and current income levels in colonized areas, reinforcing the authors’ “reversal of fortunes” argument. Even with the missing “Ugly” classification, our replication validates the main conclusions of Bruhn and Gallego (2012) and highlights the persistent influence of colonial economic structures on regional development outcomes across Latin America.

5 References

Miriam Bruhn, Francisco A. Gallego; Good, Bad, and Ugly Colonial Activities: Do They Matter for Economic Development?. *The Review of Economics and Statistics* 2012; 94 (2)

6 Figures

Figure 2: Map Displaying Areas with Good, Bad, and No Colonial Activities



7 Tables

Table 1: Regional PPP GDP per Capita across the Americas

	country object	Observations int64	Mean float64	Log_S_D float64	Minimum float64	Maximum float64	Ratio y max / y min
	Argentina 5.9%	8 - 48	2107.978 - 44267....	0.14 - 0.834	1244.53 - 26941....	2920.2 - 94900.8...	1.7 - 18.23
	Bolivia 5.9%						
	15 others 88.2%						
0	Argentina	24	11705.563	0.553	4578.247	40450.234	8.84
1	Bolivia	9	2715.144	0.395	1244.53	4222.66	3.39
2	Brazil	27	5753.916	0.576	1792.971	17595.828	9.81
3	Canada	13	44267.39	0.358	26941.535	94900.847	3.52
4	Chile	13	8728.378	0.423	4154.258	19820.189	4.77
5	Colombia	30	5868.953	0.489	2367.514	22314.58	9.43
6	Ecuador	22	5057.879	0.834	1457.74	26573.699	18.23
7	El Salvador	12	3336.25	0.3	2191	5954	2.72
8	Guatemala	8	3562.5	0.439	2100	8400	4
9	Honduras	18	2107.978	0.14	1715.8	2920.2	1.7
10	Mexico	32	8817.902	0.461	3664.333	23068.822	6.3
11	Panama	9	4335.65	0.676	1804.69	12696.03	7.04
12	Paraguay	18	4513.275	0.293	2843.19	7687.26	2.7
13	Peru	24	3984.093	0.57	1287.116	13295.491	10.33
14	US	48	32393.257	0.179	22205.598	53242.969	2.4
15	Uruguay	19	6723.318	0.231	3901.56	10527.52	2.7
16	Venezuela	19	5555.452	0.231	3496.789	9087.685	2.6

Table 2: Summary Statistics

	Outcome Variables	Observations flo...	Mean float64	S.D. float64	Minimum float64	Maximum float64
	Log PPP G... 5.9%	53.0 - 345.0	-11.42 - 19.33	0.13 - 6.66	-13.59 - 7.13	-8.14 - 29.0
	Log povert... 5.9%					
	15 others 88.2%					
0	Log PPP GDP per ...	345	8.83	0.97	7.13	11.67
1	Log poverty rate	331	2.93	0.92	0.21	4.4
2	Health Index	53	4.22	0.38	2.95	4.52
3	Log Gini	268	-0.74	0.16	-1.15	-0.46
4	Log schools per c...	317	-5.31	0.64	-7.29	-3.69
5	Log literacy rate	270	-0.14	0.13	-0.76	0
6	Log seats in lowe...	318	-11.42	1.19	-13.59	-8.14
7	Good activities d...	345	0.61	0.49	0	1
8	Bad activities du...	345	0.21	0.41	0	1
9	Mining dummy	345	0.12	0.33	0	1
10	Plantations dummy	345	0.09	0.29	0	1
11	No activities dum...	345	0.18	0.38	0	1
12	Log precolonial p...	345	0.3	2.27	-6.91	5.97
13	Average temperat...	345	19.33	6.66	-9.8	29
14	Total rainfall	345	1.26	0.94	0	8.13
15	Altitude	345	0.64	0.91	0	4.33
16	Landlocked dummy	345	0.56	0.5	0	1

Table 5: Colonial Activities and Current GDP per Capita

	(1)	(2)	(3)	(4)	(5)	(6)
egood	-0.139** (0.047)		-0.046 (0.532)	-0.024 (0.739)	-0.039 (0.588)	-0.039 (0.595)
ebad	-0.347*** (0.000)		-0.249*** (0.005)	-0.238*** (0.007)	-0.213** (0.017)	
lpopd		-0.078*** (0.000)	-0.068*** (0.000)	-0.070*** (0.000)	-0.065*** (0.001)	-0.066*** (0.000)
plantations						-0.184* (0.092)
mining						-0.235** (0.021)
temp_avg				0.001 (0.946)	-0.003 (0.882)	-0.003 (0.868)
temp2				-0.000 (0.766)	-0.001 (0.269)	-0.001 (0.265)
rainfall				-0.197*** (0.002)	-0.186*** (0.003)	-0.191*** (0.003)
rain2				0.025** (0.029)	0.021* (0.059)	0.022* (0.053)
alti					-0.062 (0.512)	-0.066 (0.490)
alti2					-0.030 (0.284)	-0.028 (0.318)
landlocked					-0.087 (0.145)	-0.084 (0.157)
N	345	345	345	345	345	345
R-squared	0.802	0.803	0.810	0.817	0.827	0.828
Coefficient: Bad - Good	-0.208	nan	-0.203	-0.214	-0.174	nan
F-test: Good = Bad p-value	0.001		0.001	0.001	0.006	
Coefficient: Plantations - Mining	nan	nan	nan	nan	nan	-0.051
F-test: Plantations = Mining p-value						0.653

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

Table 6: Colonial Activities and Current Poverty Rate

	(1)	(2)	(3)	(4)	(5)
egood	0.093 (0.214)		0.031 (0.692)	-0.007 (0.930)	0.026 (0.721)
ebad	0.297*** (0.001)		0.232** (0.013)	0.203** (0.025)	0.178** (0.045)
lpopd		0.054*** (0.004)	0.046** (0.021)	0.050** (0.010)	0.044** (0.018)
temp_avg				-0.001 (0.981)	0.014 (0.590)
temp2				0.000 (0.611)	0.001 (0.338)
rainfall				0.268*** (0.000)	0.253*** (0.000)
rain2				-0.028** (0.020)	-0.023** (0.044)
alti					0.027 (0.781)
alti2					0.059** (0.044)
landlocked					0.178*** (0.003)
Observations	331	331	331	331	331
R-squared	0.766	0.762	0.770	0.790	0.818
Coefficient: Bad - Good	-0.204	nan	-0.201	-0.21	-0.152
F-test: Good = Bad p-value	0.002		0.003	0.001	0.015

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

Table 7: Colonial Activities and Precolonial Development

	(1)
egood	0.475 (0.191)
post_egood	-1.271** (0.034)
ebad	0.824** (0.031)
post_ebad	-2.168*** (0.000)
lpopd	0.120 (0.137)
post_lpopd	-0.130* (0.069)
N	106
R-squared	0.586
Coefficient: Bad - Good	0.349
F-test: Good = Bad p-value	0.089

Standard errors in parentheses.

* p<.1, ** p<.05, ***p<.01

Table 8: Reversal of Fortunes

	(1)	(2)
lpopd	-0.065** (0.018)	0.107* (0.069)
N	284	61
R-squared	0.767	0.935
Regions in Sample	With colonial activities	Without colonial activities

Standard errors in parentheses.

* p<.1, ** p<.05, ***p<.01